Digital Mohalla — ER diagram, iteration-1 core (design step A6, schema.sql v2.3; generated from the PostgreSQL catalogue)

22 tables, 217 columns, 48 foreign keys (20 drawn as lines; the rest are marked → table in the row). Crow's foot: || exactly one, o| zero or one, o< zero or many, |< one or many.

app_user «User»

PK	id	uuid
	display_name	text
	email	text
	phone	text
	password_hash	text
	verification_level	verification_level
	platform_role	platform_role
	status	account_status
	is_synthetic	boolean
	created_at	timestamptz

user_block

PK,FK	blocker_id	uuid → app_user
PK,FK	blocked_id	uuid → app_user
	created_at	timestamptz

community «Community»

PK	id	uuid
	name	text
UQ	join_code	text
	rules	text
	boundary_polygon	jsonb
	is_active	boolean
	created_at	timestamptz

membership «Membership»

PK	id	uuid
FK	user_id	uuid
FK	community_id	uuid
	role	membership_role
	status	membership_status
	requested_at	timestamptz
	decided_at	timestamptz

locality_verification

append-only

PK	id	uuid
FK	membership_id	uuid
	method	verification_method
	outcome	locality_outcome
	geofence_signal	geofence_signal
FK,UQ	invitation_id	uuid → invitation
	permission_refused	boolean
	mock_provider_detected	boolean
	duplicate_suspected	boolean
FK	approved_by	uuid → app_user
	checked_at	timestamptz
	valid_until	timestamptz

category

PK	id	smallint
UQ	name	text
	is_restricted	boolean
	min_verification	verification_level

item «Item»

PK	id	uuid
FK	owner_id	uuid → app_user
FK	community_id	uuid
FK	category_id	smallint
	title	text
	description	text
	value_band	value_band
	deposit_amount	numeric(12,2)
	required_verification	verification_level
	status	item_status
	created_at	timestamptz

booking

EXCLUDE (item_id =, period &&) WHERE HELD

PK	id	uuid
FK,UQ	transaction_id	uuid
FK	item_id	uuid
	period	tstzrange
	status	booking_status
	created_at	timestamptz
	released_at	timestamptz

condition_report

append-only

PK	id	uuid
FK	transaction_id	uuid
	phase	condition_phase
FK	reporter_id	uuid → app_user
	reporter_role	party_role
	checklist	jsonb
	condition_agreed	boolean
	deposit_received	boolean
	reduced_evidence	boolean
	notes	text
	created_at	timestamptz

invitation

PK	id	uuid
FK	community_id	uuid → community
FK	inviter_id	uuid → app_user
UQ	code_hash	bytea
	created_at	timestamptz
	expires_at	timestamptz

borrowing_transaction «BorrowingTransaction»

projection of lifecycle_event

PK	id	uuid
FK	item_id	uuid
FK	borrower_id	uuid → app_user
FK	lender_id	uuid → app_user
	requested_period	tstzrange
	message	text
	created_at	timestamptz
	current_state	lifecycle_state
	stream_version	integer
	hold_reason	hold_reason
	due_at	timestamptz
	last_event_at	timestamptz
	closed_at	timestamptz

lifecycle_event «LifecycleEvent»

append-only · hash-chained

PK	event_id	bigint
FK	stream_id	uuid
	stream_seq	integer
FK	event_type	event_type
	schema_version	smallint
FK	actor_id	uuid → app_user
	occurred_at	timestamptz
FK	previous_state	lifecycle_state
FK	new_state	lifecycle_state
	evidence_refs	jsonb
	authorisation	text
	config_version	text
	payload	jsonb (ADR-0005)
	prev_hash	bytea
	row_hash	bytea

allowed_transition

append-only

PK	from_state	lifecycle_state
PK	event_type	event_type
PK	to_state	lifecycle_state
	allowed_roles	actor_role[]
	ref	text
	iteration	smallint

audit_log

append-only · hash-chained

PK	id	bigint
	occurred_at	timestamptz
FK	actor_id	uuid → app_user
	authorisation	text
	action	text
	target_type	text
	target_id	text
	previous_value	jsonb
	new_value	jsonb
	evidence_refs	jsonb
	outcome	text
	reason	text
	prev_hash	bytea
	row_hash	bytea

assessment «Assessment»

append-only · fields: Student 2

PK	id	uuid
FK	transaction_id	uuid
	generated_at	timestamptz
	config_version	text
	assurance_state	assurance_state
	policy_decision	policy_decision
	hold_reason	hold_reason
	posterior_mean	numeric(5,4)
	ci90_low	numeric(5,4)
	ci90_high	numeric(5,4)
	effective_evidence	numeric(8,3)
	direct_evidence	jsonb
	indirect_evidence	jsonb
	manipulation_risk	smallint
	manipulation_level	smallint
	manipulation_corroborated	boolean
	manipulation_signals	jsonb
	safeguards	jsonb
	explanation	text
	rule_fired	text

outcome «Outcome»

fields: Student 2

PK	id	uuid
FK	transaction_id	uuid → borrowing_transaction
FK	source_event_id	bigint
FK	subject_id	uuid → app_user
FK	observer_id	uuid → app_user
	subject_role	party_role
	metric	outcome_metric
	success	boolean
	occurred_at	timestamptz
FK	category_id	smallint → category
	value_band	value_band
	loan_duration	interval
	status	outcome_status
FK	flagged_by_decision_id	uuid

trust_edge «TrustEdge»

PK,FK	source_id	uuid → app_user
PK,FK	target_id	uuid → app_user
PK	role	party_role
PK	metric	outcome_metric
	weight	numeric(4,3)
	evidence_count	integer
	updated_at	timestamptz

review «Review»

fields: Student 2

PK	id	uuid
FK	transaction_id	uuid
FK	reviewer_id	uuid → app_user
FK	reviewee_id	uuid → app_user
	reviewee_role	party_role
	ratings	jsonb
	comment	text
	retaliation_flag	boolean
	similarity_flag	boolean
	created_at	timestamptz

escalation_case «EscalationCase»

PK	id	uuid
FK	transaction_id	uuid
FK,UQ	opened_event_id	bigint
	trigger_type	case_trigger
	hold_reason	hold_reason
	dispute_reason	dispute_reason
FK	raised_by	uuid → app_user
	status	case_status
	priority	smallint
	reason	text
FK	assigned_to	uuid → app_user
	info_deadline	timestamptz
	opened_at	timestamptz
	closed_at	timestamptz

moderator_decision «ModeratorDecision»

append-only

PK	id	uuid
FK	case_id	uuid
FK	decided_by	uuid → app_user
	decider_role	actor_role
	kind	decision_kind
	reason	text
	safeguards	jsonb
	resolved_outcome	jsonb
FK	reverses_decision_id	uuid → moderator_decision
	decided_at	timestamptz

case_statement

append-only

PK	id	uuid
FK	case_id	uuid
FK	author_id	uuid → app_user
	author_role	party_role
	statement	text
	created_at	timestamptz

attachment

PK	id	uuid
FK	transaction_id	uuid → borrowing_transaction
FK	condition_report_id	uuid → condition_report
FK	case_statement_id	uuid
FK	uploaded_by	uuid → app_user
UQ	storage_key	text
	content_sha256	bytea
	mime_type	text
	metadata_stripped	boolean
	created_at	timestamptz

Legend

Header colour = module: 1 Accounts & communities · 3 Borrowing lifecycle · 5 Assurance · 9 Moderation & audit · reference data

«Class» = domain-model class (A2) the table realises. PK / FK / UQ = primary, foreign, unique key. Black column = NOT NULL, grey = nullable.

→ table = foreign key not drawn as a line (mostly 'who did it' columns to app_user).

lifecycle_event (previous_state, event_type, new_state) → allowed_transition is one composite foreign key (the A3 transition table).

No column stores coordinates, an address or a CNIC (privacy scan in test_schema.py).